

Scalability & Future Plan

OptiCrop — Smart Agricultural Production Optimization Engine

Current Scalability

The architecture cleanly separates data, model, and application layers, meaning the trained model can be swapped, retrained on a larger dataset, or replaced with a more advanced algorithm without changing the Flask application code.

Planned Enhancements

- Real-time weather API integration, so rainfall/temperature/humidity can be auto-filled for a given location instead of manually entered.
- Fertilizer and irrigation guidance generated alongside the crop recommendation, building on the suitability-scoring logic already explored during development.
- A larger, regionally-specific dataset (state- or district-level) to improve robustness across India's diverse agro-climatic zones.
- A research/policy insights dashboard exposing the correlation and clustering analysis already performed in the notebook, for use by agricultural researchers and policymakers.
- Cloud deployment (e.g. Render, Railway, or Azure) to make the application publicly accessible beyond local hosting.